

# Leveraging Nursing Notes with Large Language Model Embeddings for Acute Kidney Injury Prediction

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## Abstract

**Background:** AKI affects 10-20% of hospitalized adults with high morbidity and mortality. Structured EHR models lack clinical context recorded in nursing notes before deterioration. **LLM encoders convert these narratives into dense semantic representations that complement structured features.**

**Methods:** Retrospective cohort study (NHIS Ilsan Hospital; n = 72,178; AKI 11.5%). A Transformer model processed time-series events with text embeddings from Qwen3-0.6B-Embedding. **Three configurations** compared at 0, 48, and 72h before AKI onset.

**Results:** The **multimodal** model achieved the **highest AUROC** (0.974 at 0h) and **superior AUPRC across all horizons**. AUPRC improvements were particularly notable, with +14.2% gain at 72h, underscoring clinical utility in imbalanced settings.

**Conclusion:** Integrating LLM-encoded nursing notes into time-series models **consistently improves AKI prediction**, supporting multimodal approaches for early identification of at-risk patients.

## Introduction

AKI affects 10–20% of hospitalized adults with high morbidity and mortality. Structured time-series models show strong performance but often miss clinical context preceding measurable deterioration. **Nursing notes capture what vitals and labs cannot—subjective complaints, qualitative urine changes, bedside clinical reasoning—often before structured variables reflect decline.** Yet this information remains underutilized due to documentation heterogeneity across institutions and languages.

**Objective:** To evaluate whether **integrating LLM-derived nursing note embeddings improves AKI prediction** compared to structured time-series data alone.

## Methods and Materials

**Study Cohort:** Retrospective study at NHIS Ilsan Hospital (n = 72,178; AKI 11.5%). Adult inpatients; AKI by KDIGO criteria

### Data Representation:

- Structured:** Demographics, vitals, and labs as time-stamped clinical events.
- Unstructured:** **Nursing notes** embedded via Qwen3-0.6B-Embedding (multilingual; Korean and English) and aligned to the time-series as discrete temporal events.

**Model Architecture:** ITE-Transformer, a **multimodal adaptation of STraTS**, encoding **unified sequences** of structured and text-embedded events within a shared Transformer encoder.

**Prediction Horizons:** 0, 48, and 72 h before AKI onset.

| Horizon         | Time-series     | Text            | Multimodal      |
|-----------------|-----------------|-----------------|-----------------|
| 0h (Diagnostic) | 0.9550 / 0.7734 | 0.9683 / 0.8197 | 0.9741 / 0.8579 |
| 48h             | 0.9285 / 0.7123 | 0.9459 / 0.7397 | 0.9396 / 0.7872 |
| 72h (Extended)  | 0.9144 / 0.6824 | 0.9488 / 0.7556 | 0.9530 / 0.7787 |

Table 1. Model Performance (AUROC / AUPRC)

## Results

### NHIS Ilsan Hospital (n=72,178; AKI by KDIGO)

#### 0h (Diagnostic)

- Multimodal best overall (AUROC 0.974 / AUPRC 0.858)
- Adding nursing notes: **AUPRC +0.085 over time-series alone**

#### 48h Early Warning

- Text alone achieved highest AUROC (0.946), above multimodal (0.940) and time-series (0.929)
- Multimodal retained **best AUPRC (0.787)**, **+0.075 over time-series**

#### 72h Extended Forecasting

- Multimodal: AUROC +0.039, **AUPRC +0.096 over time-series**
- Text: AUROC +0.034, AUPRC +0.073 over time-series

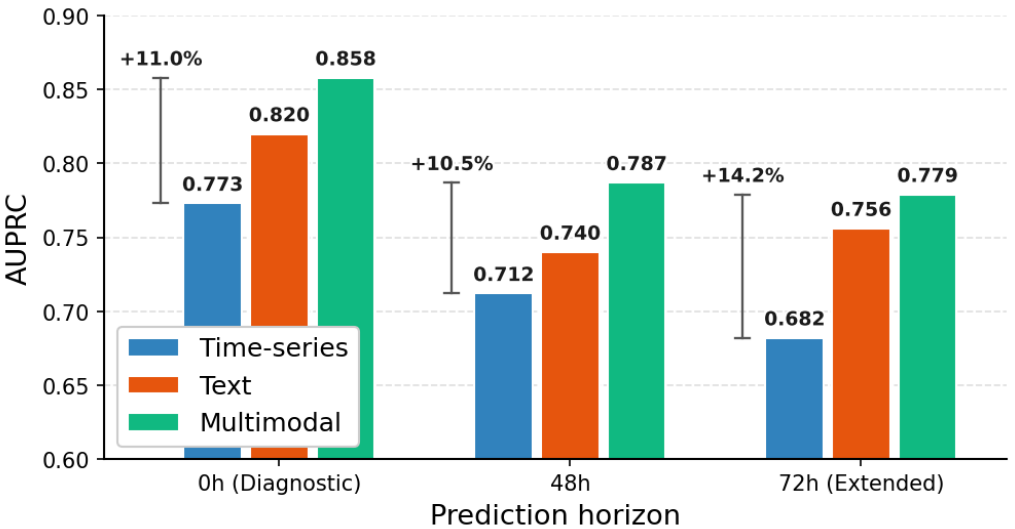


Figure 1. Precision (AUPRC)

## Discussion

### Nursing Notes as a Predictive Resource

- Incorporating notes improved prediction at all horizons, with gains widening at longer lead times
- Notes naturally reference vitals and labs within narrative context, which may partly explain strong text-only performance

### Complementary Modalities

- Multimodal consistently achieved higher AUPRC than text-only, suggesting structured data provides information not fully captured through narrative mentions

### Clinical Relevance of AUPRC

- Given class imbalance (AKI 11.5%), AUPRC better reflects real-world utility; multimodal AUPRC gain of +14.1% at 72h suggests clinically meaningful improvement

## Conclusions

- Nursing notes provide meaningful predictive value** for AKI, complementing structured time-series data across all prediction horizons
- Multimodal fusion yields the strongest performance**, supporting integration of routine nursing documentation into early-warning systems
- Limitations: single-center study; external validation and prospective evaluation are needed



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